

did_imputation and event_plot: User Guide

This document explains how to use the two R commands implemented in Tools/did_imputation

1. Setup

```
source('Tools/did_imputation.R')  
library(data.table)  
library(fixest)  
library(ggplot2)
```

2. did_imputation()

Purpose: Estimate treatment effects in staggered adoption DiD using imputation (Borusyak, Jaravel, Spiess style workflow).

Core call:

```
fit <- did_imputation(data = df, y = 'outcome', i = 'unit_id', t = 'year', ei = 'trea
```

Required arguments:

- data: data.frame/data.table
- y: outcome variable name
- i: unit id variable name
- t: time variable name
- ei: treatment timing variable name (NA for never-treated)

Important options:

- horizons = c(0,1,2,...): estimate tau by horizon
- all_horizons = TRUE: use all non-negative horizons present in data
- hbalance = TRUE: enforce balanced horizon sample
- pretrends = K: add K lead coefficients pre1...preK and run joint pretrend test

- controls = c('x1','x2'): time-varying controls
- fe = c('unit','year','state#year'): fixed effects (use # for interactions)
- cluster = 'cluster_id': cluster-robust SE target
- autosample = TRUE: auto-drop treated observations that cannot be imputed
- shift = 0: anticipation shift applied in relative-time K construction
- weights = 'w' or numeric vector: estimation weights
- wtr = NULL or c('w1','w2'): custom treated-observation weights
- sum = TRUE: interpret weighted estimand as sum instead of average
- hetby = 'group': split treatment effects by subgroup
- project = c('z1','z2'): project effects on covariates
- minn = 30: minimum effective sample size threshold

Example: dynamic effects with pretrends

```
fit <- did_imputation(
  data = df,
  y = 'y', i = 'id', t = 'year', ei = 'first_treat_year',
  horizons = 0:8,
  pretrends = 5,
  controls = c('x1','x2'),
  cluster = 'id',
  autosample = TRUE
)
```

Outputs (fit object):

- fit\$coefficients: named vector (tau*, pre*, controls)
- fit\$vcov: variance-covariance matrix
- fit\$table: estimate/SE/CI table
- fit\$Nt: treated observation counts by effect
- fit\$Nc: number of control observations
- fit\$pre_F, fit\$pre_p, fit\$pre_df: joint pretrend test
- fit\$cannot_impute: logical flag per row
- fit\$sample: estimation sample flag per row

3. event_plot()

Purpose: Plot leads/lags with confidence intervals from did_imputation and other supported models.

Core call after did_imputation:

```
pobj <- event_plot(fit)
print(pobj$plot)
```

Model inputs accepted:

- did_imputation object
- fixest or lm model
- list(b = named_coef_vector, V = vcov_matrix_optional)

How coefficients are found:

- Uses stub_lag and stub_lead patterns, where # marks the numeric horizon
- did_imputation default stubs: stub_lag = 'tau#', stub_lead = 'pre#'

Important options:

- stub_lag, stub_lead: coefficient-name patterns
- trimlag, trimlead: max lag and lead depth to display
- plottype: 'connected', 'line', or 'scatter'
- ciplottype: 'auto', 'rarea', 'rcap', or 'none'
- together = TRUE: draw leads/lags as one series per model
- shift: integer horizontal shift in event time
- perturb: offsets multiple models to avoid overlap
- savecoef = TRUE: writes __event_* vectors into caller environment
- noplot = TRUE: return plot data without drawing

Examples:

```
# Single model
p1 <- event_plot(fit, default_look = TRUE)
```

```

print(p1$plot)

# Compare two estimators
p2 <- event_plot(
  fit_bjs, fit_ols,
  stub_lag = c('tau#', 'L#event'),
  stub_lead = c('pre#', 'F#event'),
  plottype = 'scatter',
  together = TRUE
)
print(p2$plot)

```

4. Practical notes

- ei must be treatment timing, not treatment dummy.
- Never-treated units should have ei = NA.
- If imputation fails for relevant treated observations, use autosample=TRUE or adjust sample/weights.
- For fixest/lm in event_plot, always provide stubs unless coefficient names exactly match defaults.
- In this implementation, treatment-effect SEs use a clustered influence-style approximation. results may differ from Stata's iterative SE routine in edge cases.

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